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Abstract

1 Introduction

2 Preliminaries

Definition 2.1 (Positions and outcomes)**.** A terminal position (no legal moves) is a P-position; a position is an N-position if and only if at least one legal move reaches a P-position; a position is a P-position if and only if every legal move reaches an N-position, the terminal case satisfying this clause vacuously.

Definition 2.2 (The diagonal games $D(m,r)$)**.** Let $m \geq 2$ and $0 \leq r \leq m-1$. $D(m,r)$ is played on the non-negative integers. From a position $n \equiv r \pmod m$ (*squares mode*) a move subtracts any square k^2 with $k \geq 1$ and $k^2 \leq n$. From any other position (*fallback mode*) the only move subtracts m, available when $n \geq m$. Play is normal.

Definition 2.3 (Squares modulo $2m$)**.** $\mathcal{Q}(2m)$ denotes the set of residues of squares modulo $2m$. The mirror tool, proved once in general: $(2m-k)^2 = 4m^2 - 4mk + k^2 \equiv k^2 \pmod{2m}$, since $4m^2 = 2m \cdot 2m$ and $4mk = 2k \cdot 2m$. So every table is $k = 1..m$ plus mirrors, plus the 0 from $k = 2m$. Two side facts fell out of the mirrors and are used below: for odd m, $m^2 - m = m(m-1)$ has the even factor $m-1$, so $m^2 \equiv m \pmod{2m}$; for even m, $m^2 = (m/2) \cdot 2m \equiv 0 \pmod{2m}$.

mod	$\mathcal{Q}(2m)$	complement
4	$\{0,1\}$	$\{2,3\}$
6	$\{0,1,3,4\}$	$\{2,5\}$
8	$\{0,1,4\}$	$\{2,3,5,6,7\}$
10	$\{0,1,4,5,6,9\}$	$\{2,3,7,8\}$
12	$\{0,1,4,9\}$	$\{2,3,5,6,7,8,10,11\}$
14	$\{0,1,2,4,7,8,9,11\}$	$\{3,5,6,10,12,13\}$
18	$\{0,1,4,7,9,10,13,16\}$	$\{2,3,5,6,8,11,12,14,15,17\}$
22	$\{0,1,3,4,5,9,11,12,14,15,16,20\}$	$\{2,6,7,8,10,13,17,18,19,21\}$

Table 1: The engines at every modulus: $\mathcal{Q}(2m)$ with complements.

The mod-4 and mod-8 rows are the indictment written out: $\mathcal{Q}(4) = \{0, 1\}$ and $\mathcal{Q}(8) = \{0, 1, 4\}$ both sit entirely inside the triple $\{0, 1, m\}$, so their complements swallow every value any hard set will ever contain (Section 4) — the disease of $m = 2$ and $m = 4$ is visible in these two rows before any game logic starts. One eyebrow kept raised on purpose: the thinnest *odd* table is mod 18, thinned because 9 is itself a square and the rows $k = 3$ and $k = 9$ collide there — $m = 9$'s anomaly is foreshadowed in its own table.

3 The Chain Lemma

Lemma 3.1 (Chain Lemma)**.** *Fix integers $m \geq 2$ and $0 \leq r \leq m - 1$, and let c be any residue with $0 \leq c \leq m - 1$ and $c \neq r$. In $D(m, r)$, for every position $n \geq 0$ with $n \equiv c \pmod m$:*

$$n \textit{ is a } \mathrm{P}\textit{-position} \iff n \equiv c \pmod{2m}.$$

Proof. **Setup and notation.** Chain c is the set $\{c + jm : j = 0, 1, 2, \ldots\}$, i.e. all positions $\equiv c \pmod m$; call $c + jm$ rung j. Two arithmetic remarks used throughout. (i) Every rung satisfies $c + jm \equiv c \not\equiv r \pmod m$, so every position of the chain is in fallback mode: its only candidate move is subtract-m, legal exactly when the position is $\geq m$. (ii) Since $n \equiv c \pmod m$, the residue of n mod $2m$ is either c or $c + m$; and $n = c + jm \equiv c \pmod{2m} \iff 2m$ divides $jm \iff j$ is even. So the Lemma is equivalent to: rung j is P iff j is even. (All outcome computations below use Definition 2.1.)

Closure. Let $n = c + jm$ be a rung. Its only candidate move is $n \to n - m$. If $j = 0$, the move is illegal, since $n = c \leq m - 1 < m$. If $j \geq 1$, the move is legal, since $n = c + jm \geq 0 + m = m$, and it lands on $c + (j - 1)m$: the same residue c mod m, hence rung $j - 1$ of the same chain. So every legal move from a chain-c position stays inside chain c, descending exactly one rung, and the set of positions reachable from the chain is contained in the chain.

Independence. Positions outside chain c — in particular class-r positions firing squares — may well have moves that land on rungs of the chain. This cannot disturb the rungs' statuses: by the definition just quoted, the status of a position is a function only of the statuses of its options, the positions reachable from it; the definition never mentions moves arriving at a position. By Closure, every option of every rung is again a rung, so the statuses of chain c are computed entirely within chain c and would be identical if the rest of $D(m, r)$ were deleted. Incoming traffic affects only the travellers, never the rungs.

The bottom. Rung 0 is the position c, the least element of the chain, with $0 \leq c \leq m - 1$. It is in fallback mode (because $c \not\equiv r$) and its single candidate move is illegal (because $c < m$). Both facts hold equally for $c = 0$ and for $c \geq 1$ — note that for $c = 0$ no appeal to squares is needed: $0 \neq r$ already puts the position 0 in fallback mode, where $m \leq 0$ fails. So rung 0 is terminal, hence a P-position: "every legal move reaches an N" holds vacuously. This matches the claim, since $j = 0$ is even.

Alternation. Claim: rung j is P for even j and N for odd j. Strong induction on j. Base, $j = 0$: done above. Step, $j \geq 1$: assume the claim for all smaller indices. By Closure, rung j has exactly one legal move — legal precisely because $j \geq 1$, the boundary check (rung 0 has no move at all; rung 1's move is legal since $c + m \geq m$) — and it goes to rung $j - 1$. For a position with a single option x, the definition specializes: "some option is P" means x is P, so the position is N iff x is P; "every option is N" means x is N, so the position is P iff x is N. By the induction hypothesis, rung $j - 1$ is P iff $j - 1$ is even. Hence rung j is N iff $j - 1$ is even, i.e. iff j is odd; and rung j is P iff $j - 1$ is odd, i.e. iff j is even. Induction closed.

Assembly. Let $c \not\equiv r$ be any class and $n \equiv c \pmod{m}$ any position; write $n = c + jm$ with $j \geq 0$ uniquely determined. Closure and Independence license computing n's status inside the chain alone; the bottom is the base of the induction and Alternation is its step, giving: n is P $\iff$ j is even $\iff$ $jm \equiv 0 \pmod{2m}$ $\iff$ $n \equiv c \pmod{2m}$. Nothing about c was used beyond $0 \leq c \leq m-1$ and $c \neq r$, so the result holds for all $m-1$ such classes simultaneously. $\square$

4 Escape: the Clearing Lemma and its exceptions

Lemma 3.1 classifies no position of class r itself — it is silent about every $n \equiv r \pmod{m}$, and in particular does not assert those are N-positions; that claim is the subject of the present section, and only the two together give the full mod-$2m$ law.

Definition 4.1 (Window; free and hard residues; hard set)**.** By Lemma 3.1, each class $c \neq r$ is a self-contained chain whose P-positions are exactly $n \equiv c \pmod{2m}$. The window $T(m, r)$ is the set of foreign bottoms mod $2m$. Class r splits mod $2m$ into the **free residue** (r when $r \geq 1$, else m) and the **hard residue** ($r + m$ when $r \geq 1$, else 0). The free escape is immediate: 1^2 lands the free residue on the bottom directly beneath it ($r - 1$, or $m - 1$ when $r = 0$), and is legal from the smallest free member. For the hard residue, a square value $s \pmod{2m}$ lands in the window iff

$$s \in H(m, r) = \{r + 1, \ldots, m - 1\} \cup \{m + 1, \ldots, m + r\} \text{ for } r \geq 1,$$
$$\text{or} \quad \{m + 1, \ldots, 2m - 1\} \text{ for } r = 0.$$

Lemma 4.2 (Constitutional uselessness)**.** *For every m and r: $0, 1, m \notin H(m, r)$; a square $\equiv 0$ or $m \pmod{2m}$ is $\equiv 0 \pmod{m}$ and keeps a class-r position inside class r; and subtracting a square $\equiv 1 \pmod{2m}$ from the hard residue lands one below it in the top half, an* N*-position.*

Proof. The exclusions $0, 1, m \notin H(m, r)$ are read directly off the displayed form of H. A square $\equiv 0$ or $m \pmod{2m}$ is $\equiv 0 \pmod{m}$, so subtracting it keeps a class-r position inside class r; and subtracting a square $\equiv 1 \pmod{2m}$ from the hard residue lands one below it in the top half — the odd rung directly above a bottom — an N-position. $\square$

Lemma 4.3 (Clearing Lemma)**.** *Suppose some square $s \neq m$ lies in the integer interval $(r,\, m+r]$ when $r \geq 1$, or in $(m,\, 2m)$ when $r = 0$. Then the entire hard class is* N *by the single move $-s$.*

Proof. Legality: $s \leq m + r$, the smallest hard member (for $r = 0$: $s < 2m$, and the smallest positive hard member is $2m$), so the move is affordable to every member from the first. Landing: $n - s \equiv m + r - s \pmod{2m}$ is a foreign bottom — in (r, m) when $r < s < m$, in $[0, r)$ when $m < s \leq m + r$; for $r = 0$, the landing $2m - s$ lies in $(0, m)$. Combined with the free escape, class r is all N — except possibly $n = 0$, handled in Lemma 4.7. $\square$

Lemma 4.4 (The interval contains a square)**.** *For $m \geq 3$ and $1 \leq r \leq m - 1$, the interval $(r,\, m+r]$ contains a square. For $r = 0$, the interval $(m,\, 2m)$ contains a square for every $m \geq 3$ except $m = 4$.*

Proof. For $r \geq 1$: if $(r, m+r]$ contained no square, consecutive squares would straddle it, $c^2 \leq r$ and $(c+1)^2 > m + r$, so $2c + 1 > m$, hence $c \geq m/2$ for any m, hence $c^2 \geq m^2/4 > m - 1 \geq r$, since $m^2/4 - (m-1) = (m-2)^2/4 > 0$ for $m \geq 3$ — contradicting $c^2 \leq r$. The inequality degenerates at exactly one modulus, $m = 2$, where $(m-2)^2 = 0$ and the interval can indeed be empty ($(1,3]$ in $D(2,1)$ contains no square). Erratum, folded in: for $r = 0$ the interval reads $(m, 2m)$, which contains a square for every $m \geq 3$ **except** $m = 4$ — $(4,8) = \{5,6,7\}$ is square-free, and it is the lone counterexample, since the same consecutive-squares squeeze forces emptiness only for $m \leq 5$, and $m = 3$ has its 4, $m = 5$ its 9. □

Proposition 4.5 (The failure window)**.** *The hypothesis of Lemma 4.3 fails for $r \geq 1$ only if $m = b^2$ with $(b-1)^2 \leq r \leq 2b$, which is possible only for $b \leq 3$: among $m \geq 3$, exactly $m = 4$ ($r \in \{1,2,3\}$) and $m = 9$ ($r \in \{4,5,6\}$).*

Proof. With a square always present, the Clearing Lemma's hypothesis can fail for $r \geq 1$ only if the *unique* square in $(r, m+r]$ is m itself: $m = b^2$ with $(b-1)^2 \leq r$ (no smaller square intrudes) and $(b+1)^2 > m + r$, i.e. $r \leq 2b$. The window $(b-1)^2 \leq r \leq 2b$ is nonempty iff $(b-1)^2 \leq 2b$ iff $b \leq 3$. Among $m \geq 3$ that leaves exactly $m = 4$ ($b = 2$, $r \in \{1,2,3\}$) and $m = 9$ ($b = 3$, $r \in \{4,5,6\}$). With the $r = 0$ erratum, every class of $m = 4$ is condemned from the outside — no clearing square exists for any r — while $m = 9$ is healthy at every r except $4,5,6$. Squarehood of m alone is harmless: at $m = 25$ ($b = 5$) the neighboring square 36 sits only $2b+1 = 11$ above m, deep inside a 25-wide window, so it invades $(r, m+r]$ for every r; a square m isolates itself only when the modulus is small enough ($b \leq 3$) for square-gaps to outrun m. □

Proposition 4.6 (The rebels)**.** *For $(m,r) \in \{(9,4),(9,5),(9,6)\}$ the position $n = m + r$ is a* P*-position, and it is the unique* P*-position of $D(m,r)$ outside the window.*

Proof. At $(9, r)$ with $r \in \{4,5,6\}$, the smallest hard member $n_0 = m + r \in \{13,14,15\}$ can afford only the squares $k^2 \leq r$ and 9. A square $k^2 \leq r$ lands on $m + (r - k^2)$, the odd rung directly above bottom $r - k^2$ — N. The square 9 lands on r itself, the smallest free member — N, since it escapes by 1^2. Every move lands on N, so n_0 **is** P. In-game receipts: in $D(9,4)$, $13 \to 12, 9, 4$, all N; in $D(9,5)$, $14 \to 13, 10, 5$, all N; in $D(9,6)$, $15 \to 14, 11, 6$, all N. Above the rebel the class closes uniformly: the members $27+r$ $(31, 32, 33)$ are cleared by $25 \equiv 7 \pmod{18}$, which lies in all three hard sets, landing on bottoms $6,7,8$; the members $45 + r$ $(49, 50, 51)$ by $49 \equiv 13$, landing on bottoms $0,1,2$; and from $63 + r \geq 64 = (m-1)^2$ onward, the witness $64 \equiv 10 = m + 1$ lands on bottom $r - 1$ forever. So each exceptional game carries exactly one extra P-position, and the exception set is precisely the set E of Theorem 5.1. □

Lemma 4.7 (Boundary positions corrupt nothing)**.** *In $D(m,r)$: (i) if $r = 0$ then $n = 0$ is a* P*-position, and its status alters no other position's; (ii) for $(m,r) \in E$, the* P*-position $n = m+r$ alters no other position's status.*

Proof. **Corruption audits.** The $n = 0$ clause: when $r = 0$, zero is claimed by squares mode and is the unique squares-mode position that can afford no square ($k \geq 1$ forces $k^2 \geq 1 > 0$), so it is terminal, hence P — while the unamended law would file residue 0 as N. It corrupts nobody: fallback chains never visit class 0; the positions that can drop to 0 are $n = k^2 \equiv 0 \pmod{m}$, class-0 members the theorem already declares N; and the classes above escape without it — spot receipts: $10 - 9 = 1$ (P) in $D(5,0)$, $14 - 9 = 5$ (P) in $D(7,0)$. For $r \geq 1$ the clause is absorbed, since 0 is itself a foreign bottom; it earns its keep exactly at $r = 0$. The rebels, one sentence in the same mirror: a rebel corrupts nobody either — fallback chains never enter class r, and the only positions with a move onto $m + r$ are class-r members above it (a square step back into class r forces $k^2 \equiv 0 \bmod 9$, i.e. $3 \mid k$), which the theorem already declares N, so the rebel's P-ness merely hands them one more witness for what they already were. □

5 The Diagonal Theorem

Theorem 5.1 (Diagonal Law)**.** *Let $m \geq 3$ with $m \neq 4$, and let $0 \leq r \leq m-1$. Define $E = \{(9,4),(9,5),(9,6)\}$. In $D(m,r)$, a position n is a* P*-position if and only if*

$$n \equiv b \pmod{2m} \text{ for some foreign bottom } b \in \{0,1,\dots,m-1\} \setminus \{r\}, \quad \text{or} \quad n = 0, \quad \text{or} \quad (m,r) \in E \text{ and } n = m+r.$$

The law holds for both parities of m, with no thresholds and no unverified zones.

Proof. Fix a position n of $D(m,r)$ with $m \geq 3$, $m \neq 4$: either n lies in a class $c \not\equiv r \pmod m$, or n lies in class r, and in the latter case n is either a boundary position named in the statement — $n = 0$ when $r = 0$, or $n = m + r$ when $(m,r) \in E$ — or an ordinary class-r position, sitting by Definition 4.1 on either the free or the hard residue modulo $2m$; these buckets exhaust all n, and we take them in order. If n lies in a class $c \not\equiv r$, then by Lemma 3.1 (the Chain Lemma) its chain is self-contained and n is P exactly when $n \equiv c \pmod{2m}$ — precisely the foreign-bottom clause of the statement, so every such position carries its verdict. If n sits on the free residue, the move -1^2 is legal from the smallest free member onward and lands on the foreign bottom directly beneath ($r-1$, or $m-1$ when $r = 0$), a P-position *by the chain verdict of the previous sentence* (Lemma 3.1), so every free-residue position is N — and indeed the statement names no free residue among its P-positions. If n sits on the hard residue and $(m,r) \notin E$, then Lemma 4.4 together with Proposition 4.5 supplies a square $s \neq m$ in the clearing interval (the hypothesis $m \neq 4$ is load-bearing precisely here, since at $m = 4$ and only there they offer nothing for any r), and Lemma 4.3 converts that one square into the verdict: every hard-residue position is N by the single move $-s$, for both parities of m, the boundary $n = 0$ having already been set aside. If n is a class-r position with $(m,r) \in E$, Proposition 4.6 delivers the exceptional verdict: $n = m + r$ is P and is the unique off-window P-position, every other hard member being N, which combined with the free verdict above is exactly the exception clause of the statement. Finally, Lemma 4.7 certifies the boundary clauses and their harmlessness: $n = 0$ at $r = 0$ is terminal hence P, and neither its P-ness nor a rebel's disturbs any verdict already issued — every position with a move onto them is a class-r position ruled N above, and no fallback chain ever enters class r — so the classifications of the four preceding sentences stand undisturbed, and the P-positions of $D(m,r)$ are exactly those of the statement: the foreign bottoms modulo $2m$, together with $n = 0$, together with $n = m + r$ when $(m,r) \in E$. □

Remark 5.2 (The historical witness)*. The odd-m witness identity, the original route.* For odd m, $(m-1)^2 = m^2 - 2m + 1 \equiv m^2 + 1 \equiv \mathbf{m+1} \pmod{2m}$, using $m^2 \equiv m$ from Definition 2.3; and $m+1$ belongs to every hard set, landing the hard residue exactly on bottom $r-1$ ($m-1$ when $r = 0$). So the pair $(1^2, (m-1)^2)$ is a universal witness pair for all odd m at once — at the price of the threshold $n \geq (m-1)^2$. The Clearing Lemma supersedes it: a square below $m + r$ does the same job with legality free, for both parities, with no threshold. The identity remains the cleanest line in this development and the historical door to everything above.

6 The diseased moduli

Lemma 6.1 (Confinement Lemma)**.** *For $m \in \{2,4\}$ and every r: $\mathcal{Q}(2m) \subseteq \{0,1,m\}$, while $H(m,r)$ excludes 0, 1, and m. Hence no square move from a hard-residue position reaches the window: every move stays inside class r or lands on an* N*-position of a foreign chain.*

Proof. For $m \in \{2, 4\}$ and every r: $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$ — read directly off the mod-4 and mod-8 rows of Table 1 — while every hard set $H(m, r)$ excludes 0, 1, and m (Definition 4.1). Hence from any hard-residue position, every legal square move either stays inside class r (a square $\equiv 0$ or m mod $2m$ is $\equiv 0$ mod m) or lands one below on the odd rung above a bottom (a square $\equiv 1$), an N-position — never on a P-position of a foreign chain. The argument is an only-options enumeration over $\mathcal{Q}(2m)$, the mirror image of the Independence step in the proof of Lemma 3.1 — there, fallback moves cannot leave a chain; here, square moves cannot leave the class except onto N. *Consequence:* the hard class's P/N structure is decided by a self-contained internal recursion, and its very first member already defies the chain law: 3 is P in $D(2, 1)$ and 5 is P in $D(4, 1)$, where escape, if it existed, would force N. Calibration: for $m = 2$ and $m = 4$ we prove confinement — no square door into the window, so the class turns inward. We do **not** prove that these confined classes obey no eventual law of their own: "no period and no modulus up to 200 fits" is a reading from the search instrument, not a theorem, and their aperiodicity remains open. □

Remark 6.2 (Double condemnation). The two diseased moduli are exactly the two the parity-free engine cannot serve. At $m = 2$ the engine is silent: the always-a-square inequality degenerates ($(m - 2)^2 = 0$) and the clearing interval $(1, 3]$ is genuinely square-free. At $m = 4$ the engine convicts every class from the outside: the failure window covers $r \in \{1, 2, 3\}$, and the $r = 0$ interval $(4, 8)$ is square-free. Outside view and inside view agree because both are reading the same two table rows: $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$.

7 Further solved games

Example 7.1 ($D(5, 2)$, solved)**.** A position n of $D(5, 2)$ is a P-position if and only if $n \equiv 0, 1, 3,$ or $4 \pmod{10}$.

Step 1. Statement. There is an explicit threshold N_0, computed in Step 6 below, such that every position $n \equiv 2 \pmod 5$ with $n \geq N_0$ has at least one legal square move — some k with $k^2 \leq n$ — whose landing $n - k^2$ is a P-position of a fallback chain. Consequently every such n is an N-position.

Step 2. The target window. By the Chain Lemma, the fallback classes $0, 1, 3, 4 \pmod 5$ have their P-positions exactly at $n \equiv 0, 1, 3, 4 \pmod{10}$; call that set $T = \{0, 1, 3, 4\} \pmod{10}$. A landing in T is genuinely P: a number $\equiv 0, 1, 3,$ or $4 \pmod{10}$ is $\equiv 0, 1, 3,$ or $4 \pmod 5$, so it lives in a fallback chain where the lemma has jurisdiction, and its residue mod 10 matching the chain bottom's means it sits an even number of rungs up — precisely the lemma's P-rungs.

Step 3. The split. Write $n = 5q + 2$: if $q = 2t$ then $n = 10t + 2$, and if $q = 2t + 1$ then $n = 10t + 7$, so a class-2 position is $\equiv 2$ or $7 \pmod{10}$ — that is, r or $r + m$ — and the proof forks on the parity of q exactly the way the chain proof forked on the parity of j.

Step 4. Coverage line one. For $n \equiv 2 \pmod{10}$: the differences $2 - T$ are $\{2 - 0, 2 - 1, 2 - 3, 2 - 4\} \equiv \{2, 1, 9, 8\} \pmod{10}$, and intersecting with $\mathcal{Q} = \{0, 1, 4, 5, 6, 9\}$ leaves $\{1, 9\}$; take the smallest, $s = 1$, so subtracting 1^2 gives $n - 1 \equiv 1 \pmod{10} \in T$.

Step 5. Coverage line two. For $n \equiv 7 \pmod{10}$: the differences $7 - T$ are $\{7, 6, 4, 3\} \pmod{10}$, intersecting with $\mathcal{Q}$ leaves $\{4, 6\}$; take $s = 4$, so subtracting 2^2 gives $n - 4 \equiv 3 \pmod{10} \in T$. Both keystones came with a spare — $9 = 3^2$ also rescues residue 2, and $16 = 4^2$ also rescues residue 7. The corridor is wider than it needs to be.

Step 6. Legality and the threshold. The rescue squares are the concrete numbers 1 and 4. For the residue-2 class, whose members are $2, 12, 22, \ldots$, legality demands $1 \le n$, which already holds at the smallest member. For the residue-7 class, members $7, 17, 27, \ldots$, legality demands $4 \le n$, which holds at 7. So both rescues are legal for every member of their class, and the threshold collapses: $\mathbf{N_0 = 2}$, the smallest class-2 position in existence. Below N_0 there are no class-2 positions at all, so the hand-check zone is empty — verdict: vacuous. Checking the small rows directly anyway: $2 - 1 = 1$ (P) ✓, $7 - 4 = 3$ (P) ✓, $12 - 1 = 11$ (P) ✓, $17 - 4 = 13$ (P) ✓, $22 - 1 = 21$ (P) ✓, $27 - 4 = 23$ (P) ✓. Every row is rescued-anyway, rebels: none. Worth recording why it collapsed: the rescue squares 1 and 4 happen to be smaller than the smallest members of their classes. Nothing guarantees that luck at other moduli, which is exactly why the theorem was written with N_0 as a named unknown.

Step 7. Assembly. The Chain Lemma classifies the four fallback classes everywhere: P exactly at $n \equiv 0, 1, 3, 4 \pmod{10}$, N at $5, 6, 8, 9$. The Escape Theorem classifies class 2 everywhere, since $N_0 = 2$ means "past the threshold" is all of it: residues 2 and 7 (mod 10) are all N. Union the two and every residue mod 10 is accounted for, with no unexplained positions beneath any threshold.

One coincidence too pretty not to flag: the P-set of the whole game *is* the target window T — the landing pads and the answer turned out to be the same four residues (resolved in Remark 7.3).

Step 8. Scope and the forward hook. The proof secretly depended on squares mod 10 being *rich*: $\mathcal{Q}$ fills six of the ten residues, enough that both difference lists $\{2, 1, 9, 8\}$ and $\{7, 6, 4, 3\}$ caught at least one member — had $\mathcal{Q}$ missed either list entirely, one residue of class 2 would have had no escape and the whole corridor would have collapsed. Generalizing to $D(m, r)$ therefore means asking whether that richness survives at other moduli — whether squares mod $2m$ keep intersecting the difference sets $\rho - T$ for both $\rho \in \{r, r + m\}$ — and since quadratic residues thin out and organize themselves as the modulus grows, that question, not the game logic, is precisely the content of Section 4.

Remark 7.2 (Verification against the workbook). Prediction first, then the rows: residue-2 positions $(2, 12, 22)$ should be rescued by any legal square $\equiv 1$ or $9 \pmod{10}$, residue-7 positions $(7, 17)$ by any square $\equiv 4$ or $6 \pmod{10}$, and every other square should be wasted.

- $n = 2$: only move $2 - 1 = 1$ (P). Predicted rescue 1^2 present ✓, nothing else in the row to waste.
- $n = 7$: $7 - 1 = 6$ (N), $7 - 4 = 3$ (P). Rescue 2^2 present ✓; the 1^2 is wasted exactly as the congruence says ($1 \notin \{4, 6\}$ mod 10, landing $\equiv 6$).
- $n = 12$: $12 - 1 = 11$ (P), $12 - 4 = 8$ (N), $12 - 9 = 3$ (P). Primary rescue 1^2 ✓, and the spare $9 = 3^2$ makes its first appearance in the first row big enough to afford it.
- $n = 17$: $17 - 1 = 16$ (N), $17 - 4 = 13$ (P), $17 - 9 = 8$ (N), $17 - 16 = 1$ (P). Primary 2^2 ✓ and spare 4^2 ✓.
- $n = 22$: $22 - 1 = 21$ (P), $22 - 4 = 18$ (N), $22 - 9 = 13$ (P), $22 - 16 = 6$ (N). Primary 1^2 ✓, spare 3^2 ✓.

The check came back stronger than required: not only does the predicted rescue sit in every row, every wasted square is wasted for the predicted reason — its residue misses the survivor set, and every wasted landing is $\equiv 6$ or $8 \pmod{10}$, both N-residues. The congruence machinery reproduces the workbook letter-for-letter, and the 200,000 mechanical sweep stands as the bulk consistency evidence behind it.

Remark 7.3 (The $D(5, 2)$ coincidence, resolved). For any escaped game the P-set equals its window by construction: P-set = (chain P's) $\cup$ (class-r P's) $= T \cup \emptyset = T$. The equality fails exactly where escape does — the three exceptional games have P-set $= T \cup \{m+r\}$, and the diseased games' P-sets properly contain their windows.

Theorem 7.4 (Foursquare). *Consider the game on $n \geq 0$: if $n \equiv 1 \pmod 4$, subtract any positive square $\leq n$; otherwise subtract $s \in \{3, 8, 9\}$, $s \leq n$; normal play. Then n is a* P*-position if and only if $n \equiv 0, 2, 4,$ or $6 \pmod{16}$, for all $n \geq 0$ with no preperiod.*

Partition Exhaustiveness: The set of residues modulo 16 is exhaustively partitioned into claimed P-positions $\mathcal{P} = \{0, 2, 4, 6\} \pmod{16}$ and claimed N-positions $\mathcal{N} = \{1, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15\} \pmod{16}$, such that $\mathcal{P} \cup \mathcal{N} = \mathbb{Z}/16\mathbb{Z}$ and $\mathcal{P} \cap \mathcal{N} = \emptyset$.

Lemma 7.5 (Orientation Lemma). *Every state $n \equiv 0, 2, 4,$ or $6 \pmod{16}$ is even. Consequently, $n \not\equiv 1 \pmod 4$.*

Proof. For any $r \in \{0, 2, 4, 6\}$, $n = 16q + r \equiv r \equiv 0 \pmod 2$. Since n is even, $n \bmod 4 \in \{0, 2\}$, which implies $n \not\equiv 1 \pmod 4$. Thus, Square Mode never applies at any claimed P-position, and for $n \geq 9$, the full move set $\{3, 8, 9\}$ is strictly available. □

Lemma 7.6 (Odd Squares Modulo 16 Lemma). *For any odd integer $k \in \mathbb{Z}$, $k^2 \equiv 1$ or $9 \pmod{16}$.*

Proof. Express any odd integer as $k = 2m + 1$ for some $m \in \mathbb{Z}$. Squaring yields:

$$k^2 = (2m+1)^2 = 4m^2 + 4m + 1 = 4m(m+1) + 1$$

Because $m(m+1)$ is the product of two consecutive integers, it is always even. Let $m(m+1) = 2j$ for $j \in \mathbb{Z}$. Substituting gives:

$$k^2 = 4(2j) + 1 = 8j + 1$$

If j is even ($j = 2l$), $k^2 = 16l + 1 \equiv 1 \pmod{16}$. If j is odd ($j = 2l + 1$), $k^2 = 16l + 9 \equiv 9 \pmod{16}$. Thus, every odd square is congruent to 1 or 9 $\pmod{16}$. □

Proof of Theorem 7.4. (By Strong Induction on n.)

Base Cases ($0 \leq n \leq 8$). Direct manual calculation of legal move bounds and reachable states confirms:

- $n = 0$: Standard mode, no moves ≤ 0. **P-position**.
- $n = 1$: Square mode, move $s = 1^2 \to 0$ (P). **N-position**.
- $n = 2$: Standard mode, no moves ≤ 2. **P-position**.
- $n = 3$: Standard mode, move $s = 3 \to 0$ (P). **N-position**.
- $n = 4$: Standard mode, move $s = 3 \to 1$ (N). All legal moves go to N. **P-position**.
- $n = 5$: Square mode, moves $s = 1^2 \to 4$ (P), $s = 2^2 \to 1$ (N). **N-position**.
- $n = 6$: Standard mode, move $s = 3 \to 3$ (N). **P-position**.
- $n = 7$: Standard mode, move $s = 3 \to 4$ (P). **N-position**.
- $n = 8$: Standard mode, move $s = 8 \to 0$ (P). **N-position**.

Inductive Step ($n \geq 9$). Assume the theorem holds for all states $m < n$.

Part 1: Closure (Claimed P*-positions).* Let $n \equiv r \pmod{16}$ with $r \in \{0, 2, 4, 6\}$. By Lemma 7.5, $n \not\equiv 1 \pmod 4$. For all $n \geq 9$, all of $\{3, 8, 9\}$ are legal moves since $s \leq n$. We compute $r - s \pmod{16}$ across all 12 combinations:

- $r = 0 \implies 0 - 3 \equiv 13, \quad 0 - 8 \equiv 8, \quad 0 - 9 \equiv 7 \pmod{16}$
- $r = 2 \implies 2 - 3 \equiv 15, \quad 2 - 8 \equiv 10, \quad 2 - 9 \equiv 9 \pmod{16}$

- $r = 4 \implies 4 - 3 \equiv 1, \quad 4 - 8 \equiv 12, \quad 4 - 9 \equiv 11 \pmod{16}$
- $r = 6 \implies 6 - 3 \equiv 3, \quad 6 - 8 \equiv 14, \quad 6 - 9 \equiv 13 \pmod{16}$

None of the resulting residues belong to $\{0, 2, 4, 6\} \pmod{16}$. By the strong induction hypothesis, every reachable state $n - s < n$ is an N-position. Hence, n is a **P-position**.

Part 2: Escape (Claimed N*-positions).* Let $n \equiv r \pmod{16}$ with $r \in \mathcal{N}$. We display a legal move s such that $n - s \in \mathcal{P}$:

1. **Non-Square Residues ($r \in \{3, 7, 8, 10, 11, 12, 14, 15\}$):**
   - $r = 3$: $s = 3 \implies 3 - 3 \equiv 0 \in \mathcal{P}$
   - $r = 7$: $s = 3 \implies 7 - 3 \equiv 4 \in \mathcal{P}$
   - $r = 8$: $s = 8 \implies 8 - 8 \equiv 0 \in \mathcal{P}$
   - $r = 10$: $s = 8 \implies 10 - 8 \equiv 2 \in \mathcal{P}$
   - $r = 11$: $s = 9 \implies 11 - 9 \equiv 2 \in \mathcal{P}$
   - $r = 12$: $s = 8 \implies 12 - 8 \equiv 4 \in \mathcal{P}$
   - $r = 14$: $s = 8 \implies 14 - 8 \equiv 6 \in \mathcal{P}$
   - $r = 15$: $s = 9 \implies 15 - 9 \equiv 6 \in \mathcal{P}$

   Since $n \geq 9$, $s \leq n$ is satisfied in all cases.
2. **Square-Mode Residues ($r \in \{1, 5, 9, 13\}$):** By Lemma 7.6, odd squares satisfy $k^2 \equiv 1$ or $9 \pmod{16}$. Using $s = 1^2 = 1$ or $s = 3^2 = 9$:
   - $r = 1$: $s = 1 \implies 1 - 1 \equiv 0 \in \mathcal{P}$
   - $r = 5$: $s = 1 \implies 5 - 1 \equiv 4 \in \mathcal{P}$
   - $r = 9$: $s = 9 \implies 9 - 9 \equiv 0 \in \mathcal{P}$
   - $r = 13$: $s = 9 \implies 13 - 9 \equiv 4 \in \mathcal{P}$

   For $r \in \{1, 5\}$, $s = 1 \leq n$ holds for $n \geq 1$. For $r \in \{9, 13\}$, $s = 9 \leq n$ holds for $n \geq 9$.

By strong induction, every state $n \in \mathcal{N}$ can transition to a P-position $n - s < n$. Thus, n is an **N-position**. □

Remark 7.7 (No-Preperiod Remark). The periodicity of P-positions (period = 16) holds strictly and purely from $n = 0$ onward, exhibiting no initial non-periodic transient phase (preperiod = 0).

Proposition 7.8 (Odd-fallback collapse at $m = 2$)**.** (Lemma A.) *For every odd integer $a \geq 1$, the set of* P*-positions (states n with $G(n) = 0$) consists precisely of all even integers. Consequently, the emergent period is* 2 *with a residue-set size of* 1*.*

(Lemma B.) *For all odd fallbacks $a, a' \geq 1$ and all states $n \geq 0$, the Grundy values satisfy $G_a(n) = G_{a'}(n)$.*

Proof. **Setup.** We fix $r = 1$: squares fire on odd n, and the fallback rule applies on even n, making the allowed move from n to $n - a$. Let $G(n)$ denote the Grundy value of state n, with $G(0) = 0$.

*Lemma A (*P*-positions = Even Numbers).* We proceed by induction on n.

Base case ($n = 0$). Terminal position, $G(0) = 0$, so 0 is a P-position.

Inductive step. Assume for all $k < n$ that $G(k) = 0 \iff k$ is even (equivalently, $G(k) > 0 \iff k$ is odd).

Case 1: n is even. The only available move is subtraction of the odd fallback a: $n \to n - a$. Since n is even and a is odd, the destination $n - a$ is strictly odd. By the induction hypothesis, $G(n - a) \neq 0$ (if $n - a \geq 0$), or the move set is empty (if $a > n$).

- If $a > n$: the move set is $\emptyset$, so $\text{mex}(\emptyset) = 0$ and $G(n) = 0$.

- If $a \leq n$: the set of reachable values is $\{G(n-a)\}$ with $G(n-a) \neq 0$, so $\text{mex}(\{G(n-a)\}) = 0$ and $G(n) = 0$.

In both cases $G(n) = 0$. Thus all even positions are P-positions.

Case 2: n is odd. The move set consists of square subtractions $n \to n - k^2$ for all integers $k \geq 1$ with $k^2 \leq n$. Choosing $k = 1$ gives a legal move to $n - 1$. Since n is odd, $n - 1$ is even, so $G(n-1) = 0$ by the strong induction hypothesis. Thus $0 \in \{G(n-k^2) : k^2 \leq n\}$, forcing the mex to be positive, so $G(n) \neq 0$.

Therefore $G(n) = 0 \iff n \equiv 0 \pmod 2$.

Lemma B (Full Sequence Equivalence). **Even positions.** What is $\text{mex}(\{v\})$ when $v \neq 0$, and what does that force $G(n)$ to be for even n?

For even n, the reachable set is either empty (if $n < a$) or $\{G(n-a)\}$. By Lemma A, since $n - a$ is odd, $G(n-a) = v > 0$. Now $\text{mex}(\emptyset) = 0$, and $\text{mex}(\{v\}) = 0$ whenever $v > 0$. Therefore $G(n) = 0$ for every even n, regardless of the specific value of v, and regardless of whether the move exists at all. The parameter a evaporates at every even step.

Odd positions. At odd n, split the available squares k^2 by parity: which earlier positions' Grundy values enter the mex?

For odd n: if k is even, k^2 is even, so $n - k^2$ is odd; if k is odd, k^2 is odd, so $n - k^2$ is even. By Lemma A, for every odd k the landing $n - k^2$ is even with $G(n-k^2) = 0$. Thus the reachable Grundy values from odd n are $\{0\} \cup \{G(n-k^2) : k \text{ even}, k^2 \leq n\}$. The value 0 is guaranteed by the odd square moves ($k = 1, 3, 5, \dots$), and the non-zero entries depend exclusively on $G(n-k^2)$ for even k — landings on smaller odd numbers. So G at odd n is computed purely from earlier odd states, completely bypassing all even positions, and thus bypassing a entirely.

By induction on n: $G(0) = 0$; every even state yields $G = 0$ independent of a; and every odd state's value is computed from prior odd states via even-square subtractions, in which a nowhere appears. The parameter a never impacts the Grundy sequence, so all odd fallbacks collapse to the exact same game. □

8 Discovery methodology and retrodictions

Retrodiction I: the fingerprints

During the Chain Lemma verification sweeps — recorded weeks before any escape machinery existed — the ledger logged unexplained P-positions $3, 11, 23$ in $D(2,1)$ and $5, 13, 37$ in $D(4,1)$, anomalies with no mechanism attached. The Confinement Lemma now computes those games' hard residues as $3 \pmod 4$ and $5 \pmod 8$, and all six fingerprints satisfy their congruence without exception. The theory did not accommodate old data; it specified coordinates for where lawlessness is possible, and data recorded before the theory existed turns out to have been sitting on those coordinates the entire time.

Retrodiction II: the $D(9,4)$ receipt

Primary source, quoted verbatim from `DIAGONAL_WORKBOOK.md` and the raw JSON behind it:

```
| D(9,4) | mod 18 | P ≡ 0,1,2,3,5,6,7,8 | preperiod 14 | period 18 | swept 200,000 |
{"game":[9,4],"preperiod":14,"period":18,"p_res":[0,1,2,3,5,6,7,8],"pre_P":[0,1,2,3,5,6,7,8,13]}
```

This line was recorded before the rebel mechanism existed, and the mechanism now derives every number on it with zero freedom. The rebel 13 is the lone off-residue P, named in the `pre_P` field. Any eventually-periodic fit beginning at or before 13 would propagate 13's P-ness by the period, which the sweep refutes (31 is N); so every correct fit begins at 14, and the minimal preperiod is exactly rebel $+ 1 = 14$. The period is $2m = 18$, and the P-residues are the foreign bottoms $\{0, \ldots, 8\} \setminus \{4\} = \{0, 1, 2, 3, 5, 6, 7, 8\}$. Provenance note, owed and recorded: an earlier draft claimed from memory that every diagonal cell's preperiod was ≤ 1 and built a certificate on that claim; the file said otherwise all along, the claim and the certificate are struck, and this line enters the paper as the primary-source receipt that replaced them.

9 Open question

Acknowledgments and AI disclosure

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